Employee Management Module

1. Introduction

The Employee Management module is a fundamental part of the payroll information system, as it enables the administration and maintenance of up-to-date employee records within the company. This module facilitates the registration, consultation, and updating of employee data, ensuring the proper management of their employment and contractual information.

2. Objectives

- Allow the registration of employees with their personal and employment information.
- Facilitate the consultation and updating of employee records.
- Manage the employment status of each worker (active, inactive, terminated).
- Store relevant information for payroll processing and social benefits.

3. Features

3.1 Employee Registration

The system must allow the entry of employees with the following data:

- **Personal Information:** First name, last name, ID number, date of birth, address, contact details.
- **Employment Information:** Job position, department, date of hire, type of contract (fixed-term, indefinite, project-based).
- **Salary and Benefits:** Base salary, bonuses.
- **Social Security:** Health provider (EPS), pension fund, occupational risk insurer (ARL), family compensation fund.
- **Employee Status:** Active, suspended, terminated.

4. Database Model

The following is the database structure required for this module:

Table: Employee

Campo	Tipo de Dato	Descripción
id	INT (PK)	Identificador único del empleado
nombre	VARCHAR(100)	Nombre del empleado
apellido	VARCHAR(100)	Apellido del empleado
cedula	VARCHAR(20)	Número de cédula
fecha_nacimiento	DATE	Fecha de nacimiento
direccion	VARCHAR(255)	Dirección del empleado
telefono	VARCHAR(20)	Teléfono de contacto
cargo	VARCHAR(50)	Cargo en la empresa
departamento	VARCHAR(50)	Departamento del empleado
fecha_ingreso	DATE	Fecha de ingreso a la empresa
tipo_contrato	VARCHAR(50)	Tipo de contrato
salario	DECIMAL(10,2)	Salario base
eps	VARCHAR(50)	Entidad prestadora de salud
pension	VARCHAR(50)	Fondo de pensiones
arl	VARCHAR(50)	Administradora de Riesgos Laborales
caja_compensacion	VARCHAR(50)	Caja de compensación familiar
estado	VARCHAR(20)	Estado del empleado (Activo/Inactivo/Retirado)

5. Processes and Algorithms

Registration Process

- Validate that the ID number is not duplicated.
- Enter the employee’s data.
- Save the information in the database.

Update Process

- Search for the employee by ID number or name.
- Modify the necessary data.
- Confirm and update the information in the database.

Consultation Process

- Enter a search criterion (name, ID number, job position).
- Filter the results based on the entered criterion.
- Display the selected employee’s information.

6. Conclusion

The Employee Management module provides a solid foundation for the efficient handling of personnel information within the company. Its proper operation ensures data integrity and facilitates payroll processing and administrative management.